Online Resource 1

Supplementary material for the article "AXIOM: Correctness-Aware Multi-Agent Slice Admission Control with Federated Training"

Wireless Personal Communications

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S1. Overview

This supplement gives the full results of the verification runs that the main paper summarises in its section on reproducibility and fresh-seed verification. Every pass criterion was fixed in a pre-registration committed before the run it judged, and every check is reported whether it held or failed.

S2. Reruns on the original seeds

We reran eleven reported cells on their original seeds in fresh processes. AXIOM in both environments and at nineteen gNBs, its MLP variant, MAPPO, the independent DQN, the tuned rules, federated training, and the dropout cell all reproduced their stored per-seed results exactly. The rule we rely on as the Env A oracle also matched the stored oracle exactly on all 30 seeds.

The single-agent DQN is the one exception to exact reproduction. Its training code builds the network before it sets the seed, so the first seed in each batch of runs starts from an unseeded initialisation, while later seeds in the batch reproduce exactly. Five reruns of one such seed earned 7.45 to 9.67 reward per step, against 8.74 stored. Dropping the affected seeds leaves every comparison with this baseline in the same direction. The one change is in Env A at three gNBs, where AXIOM's lead moves from +0.311 to +0.241 per step but now passes the Holm correction, at $p = 0.039$ against 0.066.

S3. First fresh-seed run

We retrained every arm on fresh seeds from 3000 onwards, never used before, with 20 seeds at three gNBs, 15 in Env A at nineteen gNBs, and 10 in Env B at seven gNBs. All 26 predictions stated in advance for these fresh seeds held. In Env A at three gNBs AXIOM reached 98.2% of the oracle reward and beat the independent DQN and MAPPO at Holm-adjusted $p \leq 0.002$. At nineteen gNBs it kept 95.6% priority-slice compliance against 47.8% to 65.5% for the baselines, winning all six tests at Holm-adjusted $p \leq 0.001$.

In Env B at three gNBs AXIOM beat the static and accept-all rules in all 20 seeds and led every learned baseline at Holm-adjusted $p \leq 0.0002$. The tuned local rule again beat AXIOM, earning 9.175 against 8.782 per step, and at seven gNBs it beat every learned policy at $p \leq 0.004$. Two fresh-seed results are less favourable to AXIOM than the main campaigns. In Env A at three gNBs AXIOM won 11 of 20 seeds against the single-agent DQN but trailed it by 0.021 per step on average, at $p = 0.07$, so the pooled 40-seed lead over that baseline does not carry over to new seeds.

In Env B at seven gNBs the per-node MLP encoder beat graph attention by 1.38 reward per step, in 9 of 10 seeds at $p = 0.006$, which strengthens the finding that attention adds nothing.

S4. Second fresh-seed run

The second fresh-seed run, also specified in advance, covered Env A at seven gNBs under the ring and fully-connected layouts, at nineteen gNBs under the hex layout, and federated training, each on ten seeds. At nineteen gNBs AXIOM kept 96.5% priority-slice compliance against 47.5% to 65.2% for the baselines, winning all six tests at Holm-adjusted $p \leq 0.012$. Federation again showed no significant cost, at +0.077 with 95% CI [−1.03, 1.27].

Seven gNBs gave the weakest result among all the fresh-seed checks. AXIOM kept 96.7% priority-slice compliance and led every baseline under both layouts, and all six ring comparisons held at Holm-adjusted $p \leq 0.032$. Under the fully-connected layout, however, two weak AXIOM seeds at 57.2% and 76.9% compliance left its leads of 12.5 to 25.3 points short of significance after Holm correction over the twelve tests. The independent and single-agent DQNs also reached 80.9% and 80.6% compliance, above the 80% bound we had set, so three of the eleven checks in this run failed.

S5. Third fresh-seed run

The third fresh-seed run, again specified in advance, covered every remaining claim, and 43 of its 48 checks held. At seven gNBs under the hex layout and at nineteen gNBs under the fully-connected layout, AXIOM kept 99.0% and 95.7% priority-slice compliance and beat every baseline in all six tests at Holm-adjusted $p \leq 0.012$. A separate twenty-seed rerun of the seven-gNB fully-connected layout, reported alongside the failed ten-seed check rather than in its place, also held all six tests at Holm-adjusted $p \leq 0.007$.

Three of the five failed checks concern the baselines rather than AXIOM. On fresh seeds at seven gNBs the independent and single-agent DQNs kept 76.2% to 83.3% priority-slice compliance, above the 80% bound in three cells, so the baseline range of 56% to 76% holds only for the main campaign. The other two failures come from the load sensitivity study. At 1.02 times the nominal load two weak AXIOM seeds left it at 96.8% of the oracle reward, a gap that is not significant at $p = 0.25$, and below the single-agent DQN's mean, at 12.551 against 12.910. Under the 1.07 load and the changed weights AXIOM stayed within 0.6% of the oracle.

In Env B at seven gNBs under the fully-connected and hex layouts, the tuned local rule again beat every learned policy at $p \leq 0.004$, AXIOM was not significantly better than the static rule, and the MLP encoder again earned more than graph attention, though not significantly. Under disruption every dropout and churn cost was significant at Holm-adjusted $p \leq 0.041$, no gap between AXIOM and a baseline survived correction, and AXIOM kept a spike block precision of 0.97. Per-step DP noise lowered reward by 0.65 to 1.31 per step with no drop surviving correction, and every client took the same 17,969 noisy steps, so the quoted ε values apply unchanged. In the Env B sensitivity study the original AXIOM again earned less than the current design in all ten seeds, by 2.19 per step, while the higher learning rate and the wider encoder made no significant difference.

Finally, we reran the single-agent DQN with its initialisation seeded, through a wrapper that leaves the framework untouched, and in a two-seed test it then reproduced exactly across processes. AXIOM's lead over it in Env A at three gNBs becomes +0.254 per step at Holm-adjusted $p = 0.018$, and +0.948 on the replication seeds. In Env B the lead stays at +1.320, and at nineteen gNBs the seeded baseline collapses in 28 of 30 seeds and trails AXIOM by 32.3 points of priority-slice compliance. Across these samples AXIOM's lead over the single-agent DQN in Env A at three gNBs is therefore small and unstable, significant in some samples and absent in the fresh ones.

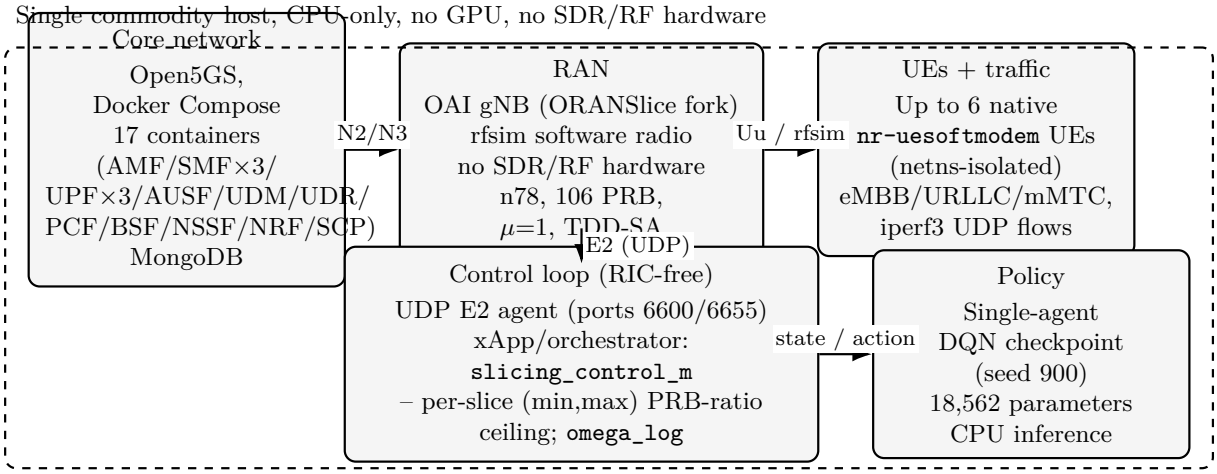


Figure 1: Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from SDR hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig.

S6. Fourth fresh-seed run

The last fresh-seed run, specified in advance, enlarged the two unstable samples, and four of its five checks held. On 40 new seeds in Env A at three gNBs AXIOM reached 98.1% of the oracle, tying it in 27 seeds, and beat the independent DQN and MAPPO at Holm-adjusted $p \leq 0.0001$. Its lead over the seeded single-agent DQN was only +0.159 per step, at $p = 0.090$. On 20 new seeds at 1.02 times the load AXIOM came within 0.5% of the oracle, but the seeded single-agent DQN again earned slightly more, 12.978 against 12.945 per step. On fresh seeds AXIOM therefore does not lead the single-agent DQN under this condition, contrary to the main sensitivity sample.

S7. Independent recomputation

A script written independently of the analysis pipeline, with its own log reader and Holm correction, recomputed from the raw logs the numbers in every results table and the key results sentences, including those of this supplement. All of its checks pass after we corrected the three rounding errors it found, as the main paper reports, and every quoted confidence interval matches a fresh bootstrap.

S8. Summary

Reruns on fresh seeds confirmed the main findings of the paper. The exceptions are a small and unstable lead over the single-agent DQN in Env A at three gNBs and baseline compliance of up to 83% at seven gNBs. In one ten-seed load sample two weak AXIOM seeds also left it 3.2% short of the oracle, a gap that a twenty-seed rerun did not reproduce.

S9. Live testbed and deployment check

The concrete rig behind every live measurement is the single-host, software-radio, O-RAN stack of Fig. 1, which runs without a near-real-time RAN Intelligent Controller (RIC). We omit the RIC because the ORANSlice fork applies slicing control directly through a lightweight UDP E2

agent, which exercises the real per-slice PRB-ratio primitive without the overhead of a separate RIC on a single commodity host.

This section reports what the real OAI testbed of Fig. 1 can and cannot establish about the offline-trained policy. AXIOM itself is not deployed on the live testbed. Its multi-gNB policy needs at least two live gNBs, and our two-gNB attempt registered both gNBs with the core but did not sustain traffic, so a live AXIOM run is left to future work. This section therefore validates the deployment pipeline that AXIOM would use, namely the E2 control path, the PRB-ratio actuation, and real-time inference of an offline-trained checkpoint. For that we run a single-agent-DQN checkpoint, seed 900, against a real gNB, real User Equipments (UEs), and real per-slice User Datagram Protocol (UDP) traffic, at one UE per slice and again at two UEs per slice.

A capacity probe at a deliberately wide-open ceiling showed all three slices serving essentially 100% of offered demand with 0.000% loss at both loads. The frozen checkpoint's decision logic runs correctly end to end, through the real E2 control loop, with real accept and reject decisions, and its signature low-priority-only blocking of mMTC reappears at both loads, which shows that the deployment pipeline works end to end. Because the rig never actually runs short of capacity, the same pattern is equally consistent with the policy's trained priority bias, so we claim admission-control behaviour under contention only in the offline shared-budget environment, not live. Two integrity issues surfaced and were fixed during these checks. A scheduler-configuration error had turned the per-slice PRB-ratio ranges into raw grants below the gNB scheduler's hard five-PRB floor, which we fixed at the root, after which direct instrumentation confirmed zero below-floor grants across more than 100,000 readings. A separate unit mismatch in the live SLA-margin diagnostic was corrected by converting backlog into a queueing-delay estimate, using Little's law, against each slice's own latency budget.

S10. Additional setup details

All learned arms share the discount factor $\gamma = 0.95$ and the training budget. The value-based arms also share the Adam optimiser, an epsilon schedule decaying from 1.0 to 0.05 at a per-episode factor of 0.985, and target synchronisation every 10 episodes. AXIOM and the DQN baselines train at learning rate 10^{-3}. The GAT encoder uses hidden and output dimension 16, feeding per-slice Q-heads of hidden dimension 32. A sensitivity study repeats the learned arms and the oracle rule in Env A on 10 seeds under three changed conditions. Two change the load, to 1.02 and 1.07 times the nominal shares, and the third lowers eMBB's priority weight from 3.5 to 2.5 and raises mMTC's from 0.3 to 0.6. A second study repeats AXIOM in Env B on 10 seeds with learning rates of 3×10^{-4} and 3×10^{-3} and with an encoder twice as wide.

The federated campaign runs in Env A on seeds 900 to 909, with a five-level noise sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$ and a replication on seeds 1000 to 1009. Each federated round runs 50 local steps under FedAvg with a gradient-clip norm of 1.0.

A validation screen checks every trained checkpoint of every learned arm before evaluation. Each checkpoint plays five greedy episodes on its own training stream, never on the held-out stream, and fails the screen if it blocks no request or if fewer than 90% of its blocks target mMTC. We report every seed whether or not it passes, and separately report a two-seed deployment protocol that trains a pair of seeds and deploys the first one to pass. The same screen applies to every learned arm, so it gives AXIOM no advantage the baselines lack. In Env B the reward-optimal policy sheds eMBB as well as mMTC, so the 90% criterion would fail good policies there. A pre-registration amendment, recorded before any result was read, therefore makes the no-blocking criterion the primary screen in Env B.

S11. Evaluation safeguard details

This shared-budget design replaces an earlier simulator that served each slice independently at about a quarter of the budget, where accepting every request was harmless. The main paper's evaluation safeguards show how a trivial accept-all baseline exposed that, and verifies that shedding mMTC is both the reward-optimal and the SLA-protecting behaviour in the current design.

The first is a trivial-policy check, in which we evaluate the simplest rules the environment admits before trusting any learned policy, namely accepting every request and shedding one slice outright. In the first version of our simulator, every slice was served independently against its own ceiling at about a quarter of the gNB's budget. There, accepting every request earned 14.45 reward per step and kept URLLC and eMBB at 100% SLA compliance, more than any learned policy, so admission control had nothing to do and policies that appeared to fail by never blocking were in fact optimal. The shared-budget simulator of the main paper removes that loophole. Over the same 30 held-out seeds, accepting every request now earns 9.89 reward per step and lets eMBB compliance fall to 56.8%, whereas shedding every mMTC request earns 12.84 and keeps URLLC and eMBB at 100%. Every result in this paper comes from the shared-budget simulator.

The third is a reproduction-versus-replication discipline, in which we recompute every headline comparison on an independent, disjoint seed sample and report both. Reproduction re-runs the same seeds, whereas replication draws fresh ones, so only replication can catch a result that is an artifact of one seed sample. It caught one such artifact in this work. A weaker replication led us to inspect step-level traces, which showed that queue backlog carried over between episodes. With only about half a PRB of headroom once mMTC is shed, a backlog built by one early transient never drained, so a single bad episode spoiled every later one for every policy, the oracle rule included. Every result in this paper comes from the corrected simulator, which empties all queues at the start of each episode.

S12. Additional results

The original AXIOM, at learning rate 10^{-4}, came slightly closer to the oracle in this environment, tying it in 39 of 40 seeds. The change to the current design therefore trades a little static-load accuracy for the ability to adapt that the time-varying environment rewards.

Three checks test where AXIOM's behaviour comes from. First, the learning rate matters, for AXIOM and its baselines alike. Every learned arm now trains at 10^{-3}, and the DQN baselines lose reward when rerun at 10^{-4}, falling to 11.477 and 11.119 per step. AXIOM depends on its rate even more in Env B. There the original AXIOM, at 10^{-4}, earns only 6.496 per step on seeds 900 to 909, 2.37 below the current AXIOM in all ten seeds, and a rate of 3×10^{-4} reaches 7.220. A rate of 3×10^{-3} and an encoder twice as wide both stay within 0.19 of the current AXIOM, with no significant difference.

FedProx on three seeds gives no consistent picture, with μ=0.01 above FedAvg and μ of 0.1 and 1.0 below it. The DQN baselines are not federated, because averaging the independent learners' weights would turn them into a parameter-sharing learner that is no longer the ablation, and the single-agent DQN is one central controller with nothing to federate.

Adding DP-SGD noise to every step lowers reward by 0.77 to 1.25 per step against centralised AXIOM at every tested noise multiplier, from σ=0.5 to σ=4. It also lowers eMBB compliance to between 74.3% and 83.0%, and two and four of the ten seeds collapse at σ of 2 and 4. None of these drops survives Holm correction at ten seeds, as Fig. 2b reflects. The formal guarantee is weak at these settings. Each client takes 17,969 noisy steps, so the zCDP accounting of the main paper gives ε of about 146,000, 37,000, 9,600, and 2,600 at δ=10^{-5} for σ of 0.5, 1, 2, and 4, far from the single-digit values that carry meaning. Per-step DP-SGD therefore delivers no

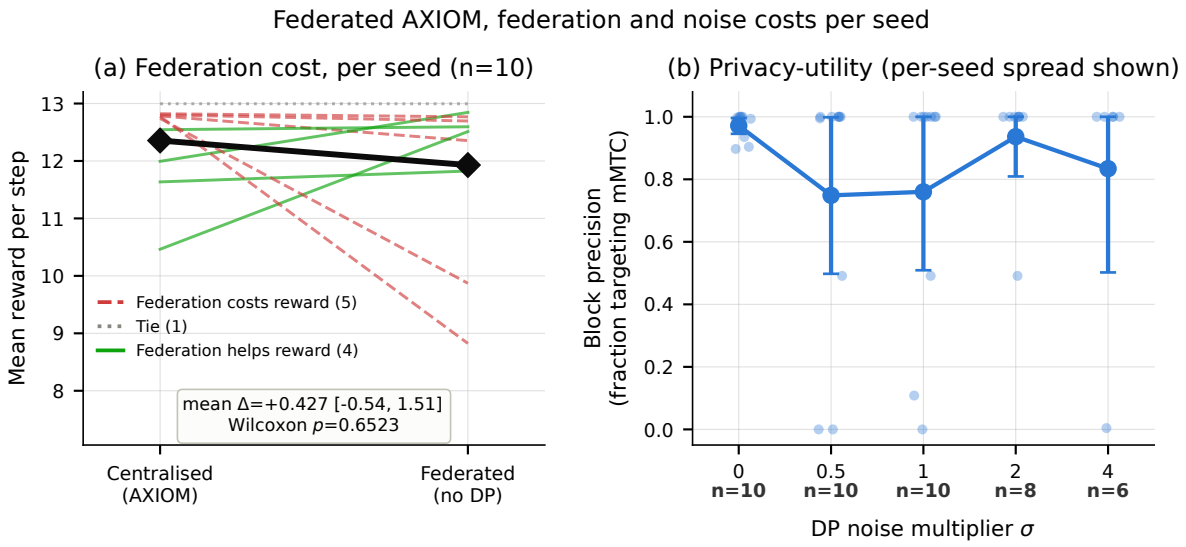


Figure 2: Federated campaign in Env A, 10 seeds. a, Paired per-seed federation cost on mean reward per step, not significant at $p = 0.65$. b, Block precision versus DP noise multiplier σ, over the seeds that block.

meaningful guarantee here. A useful guarantee would need noise on the roughly 360 per-round model updates rather than on every step, which the main paper lists as future work.

We test every policy's frozen Env A checkpoint against three mid-episode disruptions, injected at evaluation time with no retraining. A 'gNB dropout' forces one gNB to reject everything and blanks its row in the shared feature matrix for a window. A 'demand spike' briefly multiplies the arrival rate from outside the policy. An 'agent churn' swaps one gNB's decisions for a fresh, untrained copy of its own policy for a window and leaves the others intact. Churn applies to every genuinely multi-agent policy but not to the single-agent one. For dropout and churn the severity is how long the window lasts, at 10%, 30%, or 60% of the episode. For the spike it is the arrival multiplier, at 2×, 4×, or 8×. Because raw reward grows mechanically with request volume, we report a volume-normalised reward over an episode of T steps, $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t / \max(n_t, 1)$, which divides each step reward r_t by its own pending-request count n_t before averaging.

Fig. 3 plots the reward each policy keeps. Undisrupted, AXIOM earns 3.56 normalised reward, the single-agent DQN 3.65, and the independent DQN 3.32. Dropout and churn hurt every policy significantly at every severity, in 9 or 10 of 10 seeds at Holm-adjusted $p \leq 0.041$. AXIOM's cost grows from 0.46 to 3.37 across the dropout windows and from 0.38 to 2.46 across the churn windows. A dropped gNB rejects every request, including URLLC and eMBB, so its penalties dominate the episode whatever the policy does. No gap between AXIOM and any baseline survives Holm correction under any disruption. Under the two longer churn windows AXIOM keeps less reward than the independent DQN, 1.88 and 1.10 against 2.29 and 1.72, though not significantly. Demand spikes leave AXIOM's decisions largely intact, with block precision at 0.98, against 1.00 for the single-agent DQN and about 0.95 for the independent DQN. A disjoint replication on seeds 1000 to 1009 reproduces this pattern, with every dropout and churn cost significant at $p \leq 0.035$ and no significant gap between policies.

Replication first proved its worth in this campaign, on the earliest simulator. There, a ten-seed sample suggested that the independent DQN was immune to churn, since it never reads another agent's features, and a disjoint replication overturned that reading with a significant cost at every severity. In the shared-budget environment the primary sample already shows that churn hurts every policy, so the corrected reading holds.

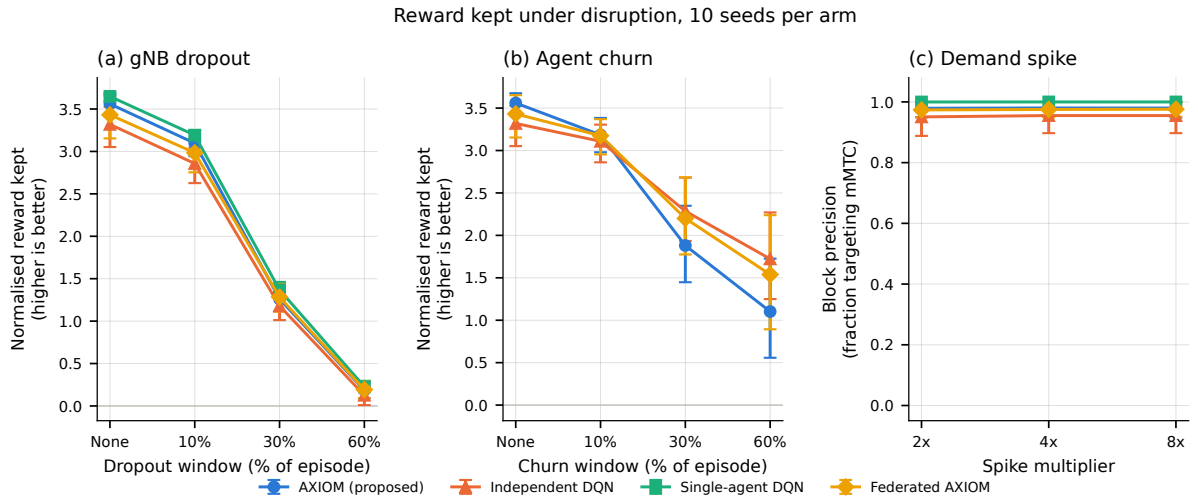


Figure 3: Disruption campaign, 10 seeds per arm, 95% CIs. a, b, Normalised reward kept under dropout and churn, with the undisrupted level at the tick marked None. c, Block precision under demand spikes.

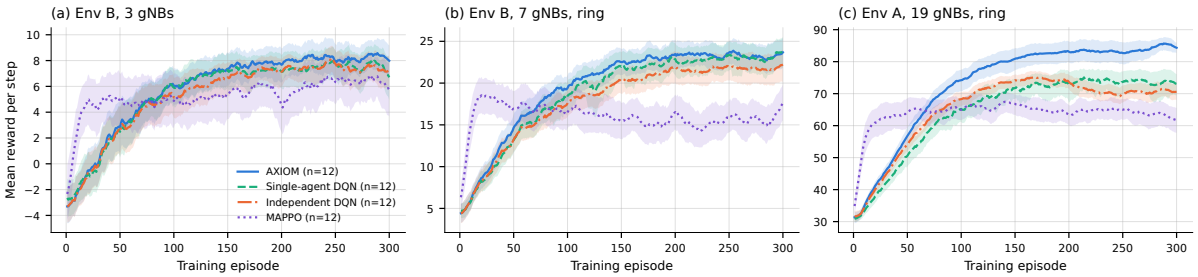


Figure 4: Training curves, mean step reward per training episode over 12 seeds with 95% bands, smoothed over 10 episodes. a, b, Env B at three and seven gNBs. c, Env A at 19 gNBs on the ring layout.

The pre-registered validation screen rarely rejects AXIOM in Env A. It fails 1 of 30 seeds at three gNBs, 1 of 45 at seven, and 5 of 90 at nineteen, including the one seed that collapses. The screen fails 89 of the single-agent DQN's 90 seeds at nineteen gNBs. It also fails many MAPPO seeds, because MAPPO also blocks eMBB, so the 90% criterion flags it. In Env B no learned policy collapsed at any size.

Fig. 4 shows how the four learned policies train. MAPPO rises fastest in the first few episodes but plateaus lowest in every cell, ending at 6.4, 16.0, and 62.9 in the three panels. The value-based arms rise more slowly but further. At nineteen gNBs AXIOM keeps improving to about 85 per step over the last 20 training episodes, while the two DQN baselines settle near 71 and 74.

Fig. 5 gives the per-layout view of the Env A scaling results that the main paper pools over layouts.

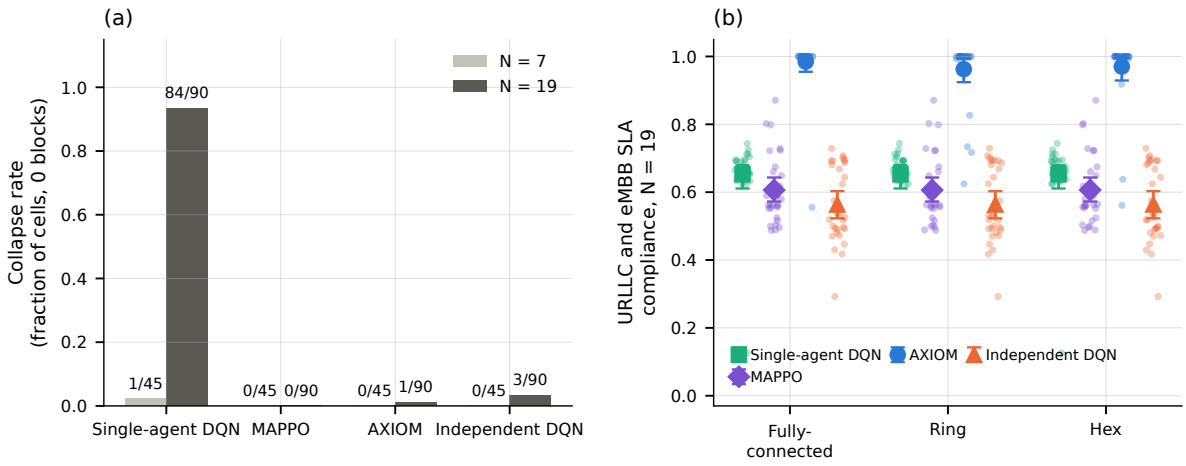


Figure 5: Env A scaling. a, Collapse rate pooled over layouts, with collapsed over total cells. b, Mean URLLC and eMBB compliance at 19 gNBs by layout, 30 seeds per cell, per-seed points behind the mean and 95% CI.